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Comments on the US AI Action Plan

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The United States must take the scientific and ethical high road as AI permeates every aspect of society. The collective consequences of AI are massive, but are dwarfed by the impact on real people's lives. There must be paths for informed, impartial input from not only the research community but also the public at large. We are members of both.

We are specifically concerned about the *quality of data used to train AI models*. Many models indiscriminately devour data of unknown and probably unknowable quality, seemingly hoping that “truth will out” in the end. Metadata, if they exist, are often ignored, to the point that it may not be known whether variables are measured, imputed, or the result of statistical modeling. There is no version control for data: who changed what, when, and why is rarely recorded. Paradata (data about the data collection process) are nearly universally absent. Ontologies for data quality do not exist.¹

But do we know that training data quality matters? In the example below, we show that it does, sometimes dramatically. And this is in a constrained scientific context with well-understood, transparent AI models—in this case, statistical classifiers. As responsible scientists, we must assume that the situation is worse for large, opaque AI models.

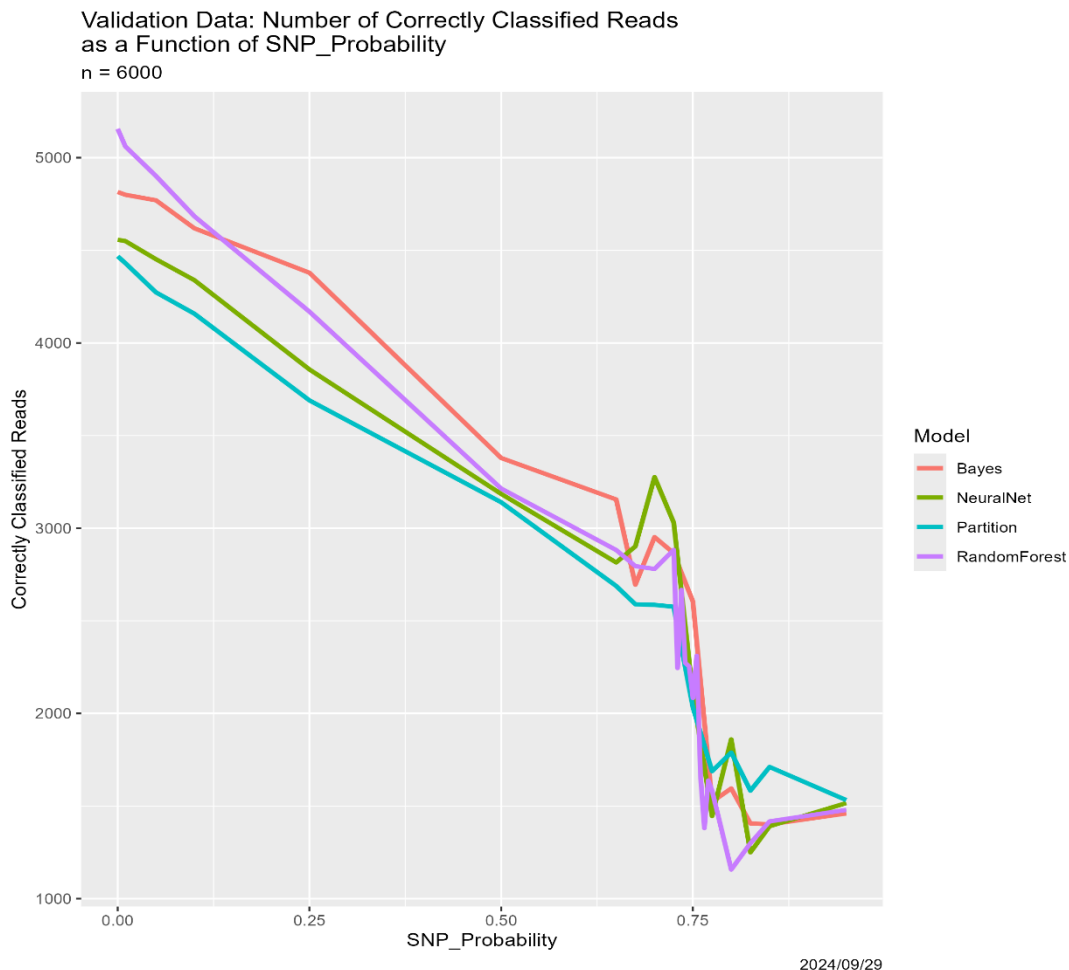
Here briefly, is one example from research we are currently conducting. The problem, which is classical in structure, is to classify short DNA reads produced by next generation sequencers as having arisen from one of three candidate genomes. This is a central step in metagenomic assembly. The three genomes, downloaded from NCBI, are an adenovirus genome, an early (2020) COVID genome and a SARS(-CoV-1) genome. The reads are simulated Illumina reads of length 101 from each of the genomes. Four classifiers were employed, a naïve Bayes classifier, a neural net, a partition model, and a random forest model, all using triplet distributions as predictors. The models are adjusted to protect

¹ One notable exception is the Total Survey Error (TSE) paradigm. See for example, P. Biemer, et al. (2017) *Total Survey Error in Practice*, Wiley, New York, to which one of us (AFK) was a contributor.

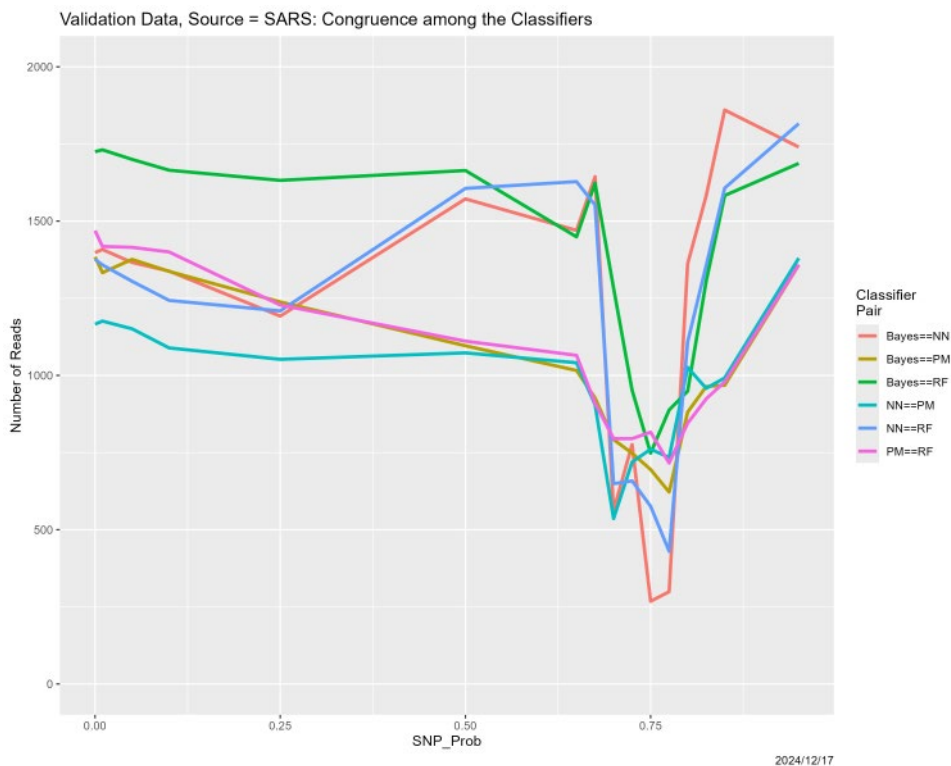
against over-fitting. The training dataset contains 5869 reads, and there is an independently generated validation dataset of size 6000. In both, the numbers of reads from each genome are approximately equal.

In general, measuring, let alone improving, data quality is expensive and/or impossible. In our experiments we employ the strategy articulated in A. F. Karr, et al. (2022), Measuring quality of DNA sequence data via degradation (2022). *PLoS ONE*. DOI: 10.1371/journal.pone.027197. But, we do know how to degrade data quality, in this instance by mimicking nature: we simulate single nucleotide polymorphisms (SNPs). Crucially, high-quality data are demonstrably more fragile than low-quality data.

Our experiments consist of increasingly degrading the training data, refitting the models and assessing effects on the validation dataset. Here is one illustrative result, showing performance of the four classifiers as the degradation (parameterized by the SNP probability) increases. The response is the number of correctly classified reads.



While the gradual decrease in performance was anticipated, the phase transition in the vicinity of SNP_Probability = .75 was not. The same happens for the congruence among the classifiers, as shown below. In this case, for SNP_Probability < .75, the congruence results from the four models being mostly, albeit decreasing, correct. For SNP_Probability > .75, they agree again, but because they are wrong in same way.



Our point, then, is straightforward: if these kinds of things can happen in simple situations, the Action Plan dares not ignore them in complex situations.

Secondary Comment: Nor, we add, do most AI models provide actionable information about uncertainties in the results they deliver. The same research mentioned above has led to surrogate measures of uncertainty for cases where the input space is a graph, which it is in the case at hand. In particular, classifier boundaries, where graph neighbors are classified differently, are complex, high-dimensional and ubiquitous. In the example discussed above, they too change rapidly at SNP_Probability = .75.

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